

Rohinish Roy

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[Portfolio](#)

EXPERIENCE

MyRohPrints 3D Design, Randolph NJ — Founder & Designer

August 2025 - PRESENT

Singlehandedly generated 50+ sales, earning Etsy Star Seller status with a perfect five-star rating.

Executed 2D and 3D design tasks using Onshape, delivering client-approved design drafts with prompt turnaround.

Oversaw financial management and daily operations to maintain workflow efficiency within budget.

FIRST Tech Challenge Team #30682 NanoGurus2.0, Morris Plains NJ— Programmer & Scouting Lead

January 2026 - PRESENT

Engineered various key robot components using laser-cut and 3D-printed materials, such as camera mounts and bumpers.

Developed autonomous pathing algorithm using the roadrunner library in Android Studio utilizing Java

Evaluated robotics teams through interpreting data and algorithmic logic in Google Sheets to determine the best alliance pair for our robot and team

Maintained a season-long engineering portfolio documenting team accomplishments.

EDUCATION

Academy for Math, Science and Engineering, Rockaway NJ

September 2024 - June 2028

Advanced Placement Classes -> Physics 1 (5), Computer Science Principles (4), Chemistry (4), Computer Science A (5)

Additional Relevant Courses: CAD1H, Fund. of Eng H, Product Development & Electricity H

SKILLS

CAD Design in Onshape & Fusion 360

Technical Drawing Drafting

FDM 3D Printer Operation & Maintenance

Laser Cutter & CNC Mill operation

PCB Design & Prototyping

Breadboarding & Microcontroller usage

Soldering

Java, C++

AWARDS/ACCOLADES

National finalist in TSA competition "Engineering Design" (2026)

Etsy Star Seller (2026)

FIRST Tech Challenge Think Award (2026)

2nd Place NJAPPT Physics Olympics (2026)

3rd Place NJAPPT Physics Olympics (2025)

3rd Place Yale Physics Olympics (2026)

PROJECTS:

Flora — *Self-Watering Flower Pot (2025)*

Desc: Flower pot which stores water and incrementally showers itself adjustably

Specifications: PETG Enclosure, custom PCB using Arduino Uno, potentiometer, resistors, and a DC 12V water pump

Functionality: Use the potentiometer to adjust the interval between pump activations, allowing you to decide when you plant needs water

OASIS— *Low-Cost Atmospheric Monitoring Module (2026)*

Desc: Atmospheric data recording module for underserved and resource-limited communities that lack access to reliable environmental monitoring infrastructure.

Specifications: Integrates an OLED Screen, a PM2.5 sensor, a MICS carbon and gas sensor, a BME sensor and an ESP 32 microcontroller into a custom made copper PCB. Enclosure made out of PETG with an underside-mounted battery for improved compactness, weight distribution, and water proofing..

Functionality: OASIS runs 24/7, relaying data it records from the atmosphere to a Granfana dashboard and an app, which tells you levels of NO, CO2, VOCs, and other atmosphere contaminants

Kick Up Game — *Fun Wall and Ball Game (2026)*

Desc: A ceiling mountable pan and tilt mechanism which flashes lights on a wall, and when a ball hits the wall in the exact position of the light, it changes the position of the light, and the cycle continues.

Specifications: The system operates on a 12V lithium-ion battery, driving a COB LED through a constant current driver, two servos across a 110° pan and 50° tilt range, an Arduino Uno controlling servo PWM on pins 9 and 10, and an ESP32-CAM communicating detections all enclosed in a custom 3D printed PLA housing ceiling-mounted approximately 1 meter from the target wall.

Functionality: Projects a repositionable circle of cool white light onto a wall one meter away, and detects when a matte black soccer ball strikes it using an ESP32-CAM running adaptive background subtraction — instantly repositioning the light to a new random location within a calibrated

DECODE Robot— *Ball collection & Shooting Robot (2025/26)*

Desc: Four wheeled competition robot which intakes up to 3 balls and shoots them with variable distance and velocity from anywhere on a field

Specifications: The robot features a mecanum drivetrain in a 14.9" × 16.4" footprint, a 6-wheel compliant intake, a 2-stage TPU sweeper transfer system, and a weighted steel flywheel beneath a servo-adjustable hood operating between 50°–70°.

Functionality: A Kalman Filter-inspired sensor fusion algorithm combining odometry and camera localization drives a gain-scheduled PIDF flywheel controller, ballistics-table trajectory lookup, automatic goal alignment, and an intake current limiter to achieve consistent 9-second artifact cycles with continuous full-field scoring.

Darksaber Prop— *Sturdy/Professional Cosplay Prop (2025)*

Desc: Star Wars inspired darksaber replica

Specifications: Its hilt was 3D printed out of PETG, and its blade was laser cut out of acrylic and MDF

Minifigure Picture Frame — *Custom Minifigure Display (2026)*

Desc: Display frame meant to hold 6 minifigures

Specifications: A CNC machined (vertical mill) block of wood designed to hold 6 minifigures. Viewing panes made out of laser cut acrylic, backing made out of PLA, backing tabs made out of laser cut MDF.

